# Ram Jet Certificates for a Calabi-Flow Singularity
# Finite Shooting, Contractive Matching, and the Geometric Boundary

Dr. Muralidhara Subbarao
Technical Report Assisted by AI Models
www.integralresearch.net
Stony Brook University
Technical Report TR-Ram-OpenAI-722-M679

October 7, 2026

**Abstract**

We analyze the pinned OpenAI manuscript on a finite-time Calabi-flow singularity on $\mathbb{CP}^{10}$, preserving its geometric reduction, positive shrinking profile, exact finite shooting architecture, and the separate ancient-inverse and compact-realization requirements. Actual private Ram jet and stability passages supply a disciplined distinction between exact coefficient identities, analytic tails, propagated residuals, and inverse bounds. A new source-specific result upgrades the manuscript's final Brouwer matching step: conditional on its stated finite ledger, the shooting map and selected small-tail branch define a contraction with constant less than 0.36 on the original parameter square. The proof controls the parameter derivative of the nonlinear shooting error and the dependence of the tail contraction on both parameters. It yields a unique matching parameter within that square and branch, certified fixed-point error bounds, and perturbation amplification at most 25/16. The finite shooting estimates are sharpened on the original square and extended to a square of 50% larger radius using the same central reference transfers and a recomputed nonlinear budget. It neither proves uniqueness among all profiles nor stability of the resulting geometric flow. Bounded exact-arithmetic tests and a successful inspected upstream certificate replay are distinguished from infinite-dimensional assertions. Contribution assessment: C3/5, a quantitatively proved conditional uniqueness and robustness upgrade at matching and a 50% finite-box extension, while the global analytic bridges remain separate requirements.

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# 1 Scope, evidence, and claims discipline

This confidential standalone report concerns manuscript M679, *A finite-time singularity of Calabi flow on projective space*, at OpenAI source commit

`adc7f1241b42e322a6451854ab7e4b4c146bf78a.`

The local catalogue rates the applicability of Ram methods L5 because finite shooting is a core existence step. That is a screening judgment, not a theorem, contribution score, or independent verification. The contribution scale here is the project's evidence-based C1–C5 scale, relative to the pinned manuscript and not a claim of historical novelty.

We use four explicit statuses. *Imported* means an assertion of the pinned manuscript or a cited private/public result. *Proved here* means an argument given in this report, with any premises stated. *Conditional extension* means a proved implication whose numerical or analytic premises are not all independently established here. *Executed test* means an actual bounded computation recorded in the local audit, not a claim that the entire theorem has been verified.

All seven substantive source sections were read: introduction, geometry, profile, parabolic, flat, compact, and certificate. The main file, preamble, cutoff figure, bibliography, README, and the three pinned verification scripts were also inspected. The original 6594-byte `certificate.py` is a separate executable appendix referenced by the main file. Its printed SHA-256 is

`263389fd7217d84d0e5b66869797fc9f7998`
`309d524a95f7ad7d3d0f538ea742.`

All three inspected pinned scripts were replayed successfully with assertions enabled. The printed certificate took 121.609 seconds. This is upstream-code replay, not independently reconstructed integration or global verification. Section 8 distinguishes these results from the additional independent bounded checks.

### 1.1 Actual private passages and their limits

The following sources were read locally, not merely through catalogue headings. They are confidential inputs, not publicly available references. Their exact paths, read ranges, and hashes belong to the freeze manifest.

1. *TR1 linear/nonlinear revision 2* [7], `sec-03-jets.tex`: full Leibniz rows, differentiated remainder, recovery theorem, nonlinear jet map, and nearby consistency/ledger passages (lines 481–950 and 1181–1500). These distinguish a polynomial-class identity from an analytic approximation and distinguish a differentiated truncation remainder from the full residual row.

2. *TR1 linear/nonlinear* [8], `stability.tex`, read in full: perturbed finite inverse, preconditioned residual-ball theorem, radii inequality, bounded coordinate transfer, and right-inverse/domain warnings. The residual-ball theorem assumes an isomorphism; a least-squares left inverse alone cannot certify a root of the full equation.

3. *Core pattern 5* [9], `TR-core-pat-5.tex`, lines 181–470: norm-controlled Neumann tails, frame reconstruction, Schur residuals, and Woodbury updates. Their hypotheses are finite-space or specified-operator hypotheses; neither a local factor nor a small interface residual is automatically a global certificate.

Only the useful mathematical mechanisms are transferred below. There is no assertion that a frame, graph reduction, or low-rank update is needed for this scalar ODE. In particular, the nonnormal transfer products cannot be replaced by their spectral radii, and the compact PDE right inverse cannot be replaced by a finite jet inverse.

## 2 The singularity mechanism in fixed conventions

### 2.1 Momentum variables, endpoints, and scalar curvature

Use $S = \operatorname{tr}_\omega \operatorname{Ric}(\omega)$ and the Fubini–Study potential $\log(1+|Z|^2)$. In complex dimension $n$, set

$$\ell = \log|Z|^2, \quad x = \Phi'(\ell), \quad v = \Phi''(\ell), \quad v_* = x(1-x), \quad w = x^{n-1}, \quad v = v_* + v_*^2 H.$$

For $H \in C^\infty[0,1]$ and $b = 1 + v_* H > 0$, the Legendre transform $u = x\ell - \Phi$ obeys

$$u'' = v^{-1}, \qquad u = u_* + \psi, \qquad u_* = x\log x + (1-x)\log(1-x), \qquad \psi'' = -H/b. \tag{2.1}$$

This proves endpoint regularity, not just interior ellipticity. Indeed, with $a = \psi'$, $q = |Z|^2 = x(1-x)^{-1}e^{a(x)}$. At zero, $x = e^{-a(0)}q + O(q^2)$ and $\Phi = C_0 + e^{-a(0)}q + O(q^2)$. At infinity put $\eta = q^{-1}$; $\Phi - \log q = C_1 + e^{a(1)}\eta + O(\eta^2)$. In the divisor chart $q = (1+\sum|W_i|^2)/|W_0|^2$, the local potential after removal of a pluriharmonic term is

$$\log B + \Psi(|W_0|^2/B), \qquad B = 1 + \sum |W_i|^2.$$

Its tangential Hessian is Fubini–Study and its normal coefficient is $e^{a(1)}/B > 0$. The potential difference from the reference is globally smooth, so the Kähler class is unchanged.

The metric eigenvalues on the affine chart are $x/q$ tangentially and $v/q$ radially. Thus $\det g = q^{-n}wv$, and for radial $f$, $\Delta f = w^{-1}\partial_x(wf_\ell)$. Direct differentiation gives

$$S = n(n+1) - K(H), \quad K(H) = w^{-1}(wv_*^2 H)'', \quad \mathcal{A}H = K(H)'', \quad \bar{S} = n(n+1). \tag{2.2}$$

The average follows from the volume density $w\,\mathrm{d}x$ and $\int K(H)w\,\mathrm{d}x = [(wv_*^2H)']_0^1 = 0$. The endpoint factor $wv_*^2 = x^{n+1}(1-x)^2$ also makes $K(H)$ smooth.

The scalar evolution is

$$H_t + (1+v_*H)^2\mathcal{A}H = 0. \tag{2.3}$$

To reconstruct precisely the Calabi flow, set $u(t) = u(t_0) + \int_{t_0}^t K(H)(s)\,\mathrm{d}s$. Then $\partial_t(1/v) = \mathcal{A}H = u_t''$, and the initial equality implies $u'' = 1/v$. Differentiating the dual at fixed $\ell$ cancels the $x_t$ terms and yields $\Phi_t = -u_t = S - \bar{S}$. An arbitrary time-dependent affine part of $u$ would introduce an unwanted dilation/constant term; the stated prescription removes that ambiguity.

### 2.2 Auxiliary sphere and flat core

Put $m = 2n+4$, $\Sigma = S^{m+5} \subset \mathbb{R}^m \times \mathbb{R}^6$, and $x = |z|^2$. Smooth $O(m)\times O(6)$ invariants correspond exactly to $C^\infty[0,1]$ functions; even radial smoothness at each collapsed orbit supplies smoothness in $x$ or $1-x$. The pushforward volume is $x^{n+1}(1-x)^2\,\mathrm{d}x$. On invariants,

$$L = \tfrac{1}{4}\Delta_\Sigma = x(1-x)\partial_x^2 + (n+2-(n+5)x)\partial_x, \qquad \mathcal{A} = (-L+n+3)(-L+2(n+2)). \tag{2.4}$$

One verifies the factorization on orthogonal polynomials: the eigenvalues of $L$ and $\mathcal{A}$ in degree $j$ are $-j(j+n+4)$ and $(j+1)(j+2)(j+n+2)(j+n+3)$. Symmetry follows by integration by parts with the above weight, retaining the vanishing boundary terms. Polynomial jets through order four then imply equality of differential expressions.

The flat operator is

$$\widehat{\mathcal{A}} = [x\partial_x^2 + (n+2)\partial_x]^2 = x^2\partial_x^4 + 2(n+3)x\partial_x^3 + (n+2)(n+3)\partial_x^2 = \tfrac{1}{16}\Delta_{\mathbb{R}^m}^2.$$

For $n = 10$ this means $m = 24$, $\Sigma = S^{29}$. These auxiliary dimensions must not be confused with the geometric dimension ten. With $D = x\partial_x$ and $g = 1 + xF$,

$$g_t = -g^2x^{-2}P(D)(g-1), \quad P(D) = (D-2)(D-1)(D+9)(D+10).$$

Suppose a positive profile $G$ satisfies

$$Q(y\partial_y)(G-1) = \tfrac{y^2}{2}(G^{-1} - c^{-1}), \qquad Q(D) = (D-1)(D+9)(D+10). \tag{2.5}$$

Apply $D-2$ to obtain $P(D)(G-1) = -y^2DG/(2G^2)$. For

$$\rho = \sqrt{-t}, \quad y = x/\rho, \quad h(y) = (G(y)-1)/y, \quad F = \rho^{-1}h(x/\rho), \quad t < 0, \tag{2.6}$$

$\partial_t G(x/\rho) = DG/(2\rho^2)$, proving $F_t + (1+xF)^2\widehat{\mathcal{A}}F = 0$. Smoothness at zero uses $h(y) = \int_0^1 G'(sy)\,\mathrm{d}s$.

### 2.3 What actually forces finite-time breakdown

The compact multiplier of this flat core is exactly

$$1 + v_*F = x + (1-x)G(x/\rho). \tag{2.7}$$

Thus two-sided positive bounds for $G$ preserve positivity of the approximation. This is not uniform metric equivalence to Fubini–Study: the map from $x$ to the fixed affine chart changes with time.

If an exact compact solution $H = F + V$ exists on $(-T, 0)$, stays positive, and $V(t,0) = O(\rho^{-1+\delta})$ for some $\delta > 0$, then

$$S(t,p) = 110(1 - H(t,0)) = -110G'(0)(-t)^{-1/2} + O(1 + (-t)^{(-1+\delta)/2}). \tag{2.8}$$

The point $p$ is fixed: $q = 0$ corresponds to $x = 0$ under every affine normalization. Choosing the smooth slice at $-T/2$ and translating time gives a flow up to $T_* = T/2$. Smooth uniqueness identifies it with the unrestricted Calabi flow, and divergent scalar curvature prohibits any smooth extension across $T_*$. This implication is elementary once the compact solution exists; the existence is the major analytic bridge, not an output of finite shooting.

# 3 The source profile and its finite certificate

## 3.1 Parameters and origin germ

Write

$$c = (1+d)^{-1}, \quad s = (27/16)^{1/3}(y/c)^{2/3}, \quad G = c(1+p).$$

Then (2.5) is

$$\mathcal{E}_d(p) = p''' + \frac{30}{s}p'' + \frac{B}{s^2}p' + \frac{C}{s^3}(p-d) + \frac{p}{1+p} = 0, \quad B = 751/4,\ C = -1215/4. \tag{3.1}$$

For example $Q(D) = D^3 + 18D^2 + 71D - 90$ checks the substitution. The source square and affine parameter map are

$$\begin{gathered} R = 5 \cdot 10^{-10}, \quad \mathcal{Z} = [-R, R]^2, \quad S_0 = 56, \\ \begin{pmatrix} k \\ d \end{pmatrix} = \begin{pmatrix} -.094617116666041 \\ .400682721856639 \end{pmatrix} + 10^{-8} \begin{pmatrix} 25190 & -63731 \\ 721855 & 359157 \end{pmatrix} z. \end{gathered} \tag{3.2}$$

Here and below every terminating decimal is rational. The row displacements are at most $4.44605 \cdot 10^{-13}$ and $5.40506 \cdot 10^{-12} < .011R$.

Let $r = s^{3/2}$ and $p = d + kr + \sum_{i \geq 2} a_i r^i$. For $b = (1+p)^{-1}$ the exact coefficient recurrence is

$$a_i = \frac{b_{i-2} - \mathbf{1}_{i=2}}{(27/8)(i-1)(i+9)(i+10)}, \quad i \geq 2. \tag{3.3}$$

The source proves joint holomorphy for $|r| < 3$, $|k - k_0|, |d - d_0| \leq .1$ by a Banach-algebra contraction. In the norm $\sum |a_i| 3^i$, the nonlinear tail ball has radius .2, $3|k| < .585$, $|1+d| > 1.30$, and the inverse has norm at most 2. The image radius is below $27/440$ and the Lipschitz constant below $36/440$. In particular $\sum_{i \geq 1} |a_i| 3^i < .8$, and each of $p(1), p'(1), p''(1)$ is bounded by 2 on this polydisc. This also proves real positivity before $s = 1$, not merely formal solvability.

The first free slope gives

$$G'(0) = \frac{\sqrt{27}}{4}k < 0, \qquad a_{\text{curvature}} = -\frac{110\sqrt{27}}{4}k > 0. \tag{3.4}$$

The constant $c$ cancels from this relation, a useful normalization check.

## 3.2 Reciprocal and tangent jets: what is exact

For $z(\theta) = \sum z_i \theta^i$ and $b = 1/z$,

$$b_0 = z_0^{-1}, \qquad b_i = -z_0^{-1} \sum_{t=1}^{i} z_t b_{i-t}, \tag{3.5}$$

$$\dot{b}_i = -z_0^{-1} \left( \sum_{t=0}^{i} \dot{z}_t b_{i-t} + \sum_{t=1}^{i} z_t \dot{b}_{i-t} \right). \tag{3.6}$$

These follow by differentiating $zb = 1$ in the polynomial coefficient ring and then in the parameter. In particular $\dot{b}_0 = -\dot{z}_0/z_0^2$ is essential. Truncation gives an exact identity modulo $\theta^{N+1}$, not a polynomial reciprocal identity as functions. The analytic remainder still has to be bounded. This is the precise connection with the private Leibniz and nonlinear jet passages: do not freeze coefficient derivatives, delete the constant-state variation, or declare closure of the rational nonlinearity.

At an ordinary step use inverse block length $h$, center $s_c$, $\theta = h(s - s_c)$, $\mathfrak{r} = hs_c + \theta$, $z = 1 + p$, and $E = 1 + d$. The scaled equation is

$$4\mathfrak{r}^3 z''' + 120\mathfrak{r}^2 z'' + 751\mathfrak{r} z' - 1215(z - E) = 4h^{-3}\mathfrak{r}^3(z^{-1} - 1). \tag{3.7}$$

Primes here mean $\theta$ derivatives. To preserve the actual code detail, set $\mathfrak{R} = 8hs_c$, $F_i = b_i - \mathbf{1}_{i=0}$, and negative-index coefficients to zero. The update is

$$\begin{aligned} 4(i+1)(i+2)(i+3)\mathfrak{R}^3 z_{i+3} = 4h^{-3}\sum_{a=0}^{3} c_a \mathfrak{R}^{3-a} F_{i-a} \\ - (96i + 960)\mathfrak{R}^2(i+1)(i+2)z_{i+2} \\ - 64\mathfrak{R}(i+1)(12i(i-1) + 240i + 751)z_{i+1} \\ - 512(4i(i-1)(i-2) + 120i(i-1) + 751i - 1215)z_i \\ - 1215 \cdot 512E\mathbf{1}_{i=0}, \qquad (c_0, c_1, c_2, c_3) = (1, 24, 192, 512). \end{aligned} \tag{3.8}$$

The four variations differentiate this entire expression, including $\dot{E} = (0, 0, 0, 1)$. Coefficients through degree 46 are computed, then the state and two derivatives are evaluated at $\theta = 1/8$.

### 3.3 Integer intervals, complex disks, and physical units

The certificate uses integer pairs representing $[a/Q, b/Q]$ with $Q = 2^{256}$. Multiplication uses the minimum and maximum of all four endpoint products with outward floor/ceiling rounding; reciprocals of same-sign intervals use $(\lfloor Q^2/b \rfloor, \lceil Q^2/a \rceil)$. Midpoint selection is not enclosure: it incurs at most half the width plus $Q^{-1}$.

The schedule has 32 blocks of each length $1/32, 1/16, 1/8$, covering $[1, 2], [2, 4], [4, 8]$, then 192 blocks of length $1/4$ to 56. Each block has eight steps. At a step start the certified bounds are $-.76 < p < .4$, $|p_\theta| < .06$, $|p_{\theta\theta}| < .01$, and every tangent entry in the top three rows is below 6. On $|\theta| \le 1/2$, $hs_c \ge 32$, a displacement bootstrap of .1 gives $|\mathfrak{r}| \ge 31.5$ and $|1 + p| > .14$. The Gronwall inequality

$$H_2 \le e^{.954/2}\left[.01 + \tfrac{1}{2}\{.190(.06 + H_2/2) + .011(1.3) + .015625(.86/.14)\}\right]$$

implies $H_2 < .13$, and displacement is below $.04625 < .1$. The scaled Jacobian norm is below 2.1; hence tangent entries are below $6e^{1.05} < 32$. Cauchy's estimate for each state/tangent entry after degree 44 gives $(128/3)4^{-45} < 48\,4^{-45}$. Differentiating a truncated scalar polynomial alone would not justify this allowance; the disk bound is on the derivative entries themselves.

The real substep displacement is $.008515625 < .01$. Lower starting values reduced by a strictly outward .01 give positive block margins $m_j$. Scaled transfers must be converted using $S_j = \mathrm{diag}(1, \ell_j, \ell_j^2, 1)$:

$$M_j = S_j^{-1} T_j^{\mathrm{scaled}} S_j. \tag{3.9}$$

The physical full norm is not bounded by 4.3. Only the first row and the third column have the needed factors at most one under this conjugation, giving both of their norms at most 4.3 for every ordered pair of points within one block. These are exactly the two actions needed for the nonlinear forcing and the value bootstrap.

### 3.4 The finite ledger is a substantive hypothesis

Let $U_i$ be rounded reference jets, $M_j$ exact physical transfers, $\widetilde{M}_j$ their rounded counterparts, and $Z$ the two origin variation columns with the exact $d$ row appended. Denote forward products by $T_{i,j}$ and $\widetilde{T}_{i,j}$, and their leading $3 \times 3$ blocks by superscript (3). The source asserts

$$m_j > .23, \quad \|Z\| < .02, \quad \max_{i \geq j} \|\widetilde{T}_{i,j}\| \leq 8500,$$
$$\max_i \sum_{j<i} \|\widetilde{T}^{(3)}_{i,j+1}\| \leq 570000, \qquad \max_i \sum_{j<i} \frac{\ell_j}{m_j^3} \|\widetilde{T}^{(3)}_{i,j+1}\| \leq 246000,$$
$$\max_i \|\widetilde{T}_{i,0} Z\| \leq 2.7. \tag{3.10}$$

All norms are maximum row-sum norms. Reference jumps are below $10^{-19}$ and each transfer entry error below $10^{-15}$. Expanding products around the rounded factors gives

$$\|T_{i,j} - \widetilde{T}_{i,j}\| \leq 8500[(1 + 4 \cdot 10^{-15} 8500)^{288} - 1] < .001. \tag{3.11}$$

Thus exact product norms are below 8501, the unweighted and weighted sums below 580000 and 250000, and $\|T_{i,0} Z\| < 2.72$. Products occur in the order $M_{i-1} \cdots M_j$; reversing them changes the problem. This is the private perturbation/Schur lesson in concrete form: local errors and jumps must be transported, not merely summed.

The origin tail after degree 60 is bounded by

$$E_{60} = .8(3 \cdot 61/2)^2 3^{-61} < 5.267 \cdot 10^{-26}.$$

Cauchy differentiation contributes at most $.12 E_{60}$ to the two variation-column row norm. The nonlinear parameter remainder has bound

$$E_{\text{par}} = 2[(1 - .11R)^{-2} - 1 - 2(.11R)] < 1.816 \cdot 10^{-20}.$$

With midpoint widths and grid error the full initial state remainder is below $2 \cdot 10^{-19}$ and its fourth component is zero.

For $u = p - p_{\text{ref}}$ and $a = 1 + p_{\text{ref}}$ the exact nonlinear remainder in the physical third-component equation is

$$N_a(u) = \frac{u^2}{a^2(a+u)}. \tag{3.12}$$

Under $|u| \leq 14R$, $a \geq m_j$, it is at most $1.001(14R)^2/m_j^3$. Variation of constants and the third-column bound give the endpoint remainder

$$\begin{aligned} \|e_i\| &\leq 580000\, 10^{-19} + 8501(2 \cdot 10^{-19}) + 4.3(250000)1.001(14R)^2 \\ &= .1055747504R < .11R. \end{aligned} \tag{3.13}$$

The first-row bound in a partial block improves the value bootstrap: $|u|/R < 4.3(2.72 + .11) + 4.3(1.001)14^2 R/.23^3 < 12.169035 < 14$. The lower bound $1 + p > .23 - 14R > 0$ and a linear Gronwall estimate for $p', p''$ prevent any finite-time escape on $[1, 56]$.

## 4 Tail construction and the original matching step

The asymptotic polynomial is $p_a = \sum_{j=1}^{4} d_j s^{-3j}$ with

$$\begin{aligned} a = d_1 &= Cd, & b = d_2 &= a^2 + 567a, \\ d_3 &= 2ab - a^3 + (2025/4)b, & d_4 &= b^2 + 2ad_3 - 3a^2b + a^4 + (567/2)d_3. \end{aligned} \tag{4.1}$$

For $L(\nu) = \nu(\nu-1)(\nu-2) + 30\nu(\nu-1) + B\nu + C$, the values at $-3, -6, -9, -12$ are $-567, -2025/4, -567/2, -243/4$. They give exact cancellation through $s^{-12}$ in (3.1). The source bounds

$$|d_j| < (122, 54200, 12500000, 5100000000)_j, \quad |\partial_d d_j| < (304, 100000, 65000000, 100000000000)_j.$$

These imply $|p_a| < .001$, $\|\partial_d(p_a, p_a', p_a'')(56)\| < .01$, and a residual $\|\mathcal{E}_d(p_a)\|_\infty < 3 \cdot 10^{-13}$ on $s \geq 56$. The residual estimate retains the nonlinear powers beyond degree four: their positive coefficient bounds are $8 \cdot 10^{-14}$, $2 \cdot 10^{-14}$, $3 \cdot 10^{-15}$ for powers two, three, four, plus $.001^5/(1 - .001)$ and the remaining linear term.

Define

$$D_0(\lambda) = \lambda^3 + \frac{15}{56}\lambda^2 + \frac{B}{2 \cdot 56^2}\lambda + 1 + \frac{C}{2 \cdot 56^3}.$$

Its roots are $\lambda_0 \in (-1.08659, -1.08655)$ and conjugates $\lambda_1, \lambda_2$, with $\Re\lambda_1 \in (.40933, .40938)$, $\Im\lambda_1 \in (.8671, .86721)$. For $v_i = (1, \lambda_i, \lambda_i^2)^t$ write the residual jet at 56 as $qv_0 + \xi v_1 + \bar{\xi} v_2$. The exact real projection $\Gamma$ is approximated by

$$\Lambda = 10^{-5} \begin{pmatrix} 30756 & -27384 & 33447 \\ 34622 & 13692 & -16724 \\ -2925 & -34040 & -28850 \end{pmatrix}. \tag{4.2}$$

The entrywise error is below .001, hence norm error below .003, and row norms of $\Gamma$ are below .92, .68, .68. These follow from the exact root signs, Vieta identities, and projection rows $(\lambda_j\lambda_k, -\lambda_j - \lambda_k, 1)/D_0'(\lambda_i)$.

For $b_i = 1/D_0'(\lambda_i)$, the bounded solution of $D_0(\partial_s)V = j$ with prescribed stable coordinate $q$ is

$$\begin{aligned} (V, V', V'')^t = {} & qe^{\lambda_0(s-56)}v_0 + b_0 v_0 \int_{56}^{s} e^{\lambda_0(s-u)} j(u)\,\mathrm{d}u \\ & - \sum_{i=1}^{2} b_i v_i \int_{s}^{\infty} e^{\lambda_i(s-u)} j(u)\,\mathrm{d}u. \end{aligned} \tag{4.3}$$

This gives $\|V^{(i)}\|_\infty \leq |\lambda_0|^i|q| + M_i\|j\|$, where $(M_0, M_1, M_2) < (1.96, 1.92, 1.88)$ and the nonreal initial coordinate response to $j$ has norm below .82.

Writing $N(p) = p^2/(1+p)$, the source fixed-point equation is

$$\begin{aligned} j = {} & -\mathcal{E}_d(p_a) - (30/s - 15/56)\,V'' - \left(B/s^2 - B/(2 \cdot 56^2)\right) V' \\ & - \left(C/s^3 - C/(2 \cdot 56^3)\right) V + N(p_a + V) - N(p_a). \end{aligned} \tag{4.4}$$

On $|q| \leq .26R$, $\|j\| \leq .3R$, $|V| \leq .848R$ and $|p_a + V| < .001$. The source obtains contraction constant .57, homogeneous $q$ coefficient .36, and image radius $(.0006 + .36 \cdot .26 + .57 \cdot .3)R = .2652R < .3R$. Consequently a unique tail exists in this source ball, with $|\xi_{\text{tail}}| < .246R < .25R$ and $p, p', p'' \to 0$.

The final finite assertions are, with $A_* = U_N - P_a(d_0)$ and $Y = (\widetilde{T}_{N,0}Z)_{\text{top }3}$,

$$\|A_*\| < .04R, \quad \|\Lambda A_*\| < .006R, \quad \|(\Lambda Y)_1\| < .116, \quad \|(\Lambda Y)_{2,3} - I_2\| < .001. \tag{4.5}$$

Propagating (3.13), projection error and $P_a$ variation gives the explicit source totals

$$|q_{\text{shot}}| \le .2315396R < .25R, \qquad \|\xi^{\mathbb{R}}_{\text{shot}} - z\| \le .0901084R < .13R, \tag{4.6}$$

where $\xi^{\mathbb{R}} = (\Re\xi, \Im\xi)$. The source map

$$\mathcal{B}(z) = z - \xi^{\mathbb{R}}_{\text{shot}}(z) + \xi^{\mathbb{R}}_{\text{tail}}(q_{\text{shot}}(z), d(z)) \tag{4.7}$$

maps $\mathcal{Z}$ into $.38\mathcal{Z}$, and Brouwer supplies a fixed point. Matching all three jets and ODE uniqueness joins the two pieces.

The source's all-order asymptotics are an additional argument. Formal coefficients alone are insufficient. For $W = p - p_J$, the divided difference gives $W''' + W = -e_J - a_2W'' - a_1W' - a_0W$, with $a_i \to 0$. The Green operator for $\partial_s^3 + 1$ has bounded norm $5/3$; in the weight $s^N$, for $s \ge S \ge 2N$, its norm is at most 2. A contraction in both spaces, with the actual stable coordinate at $S$, identifies the bounded remainder with the weighted remainder. Differentiating and then using a deeper truncation yields $\partial_s^\ell(p - p_J) = O(s^{-3(J+1)-\ell})$ for every finite $J, \ell$. This implies a differentiable expansion for $G$ in powers of $y^{-2}$, and smooth extension in $y^{-2}$ at infinity. No convergence of the infinite formal expansion is asserted.

## 5 New finite-box refinement and enlargement

This refinement was exact-checked and independently reviewed; its full argument is given here. It changes only the finite shooting proposition. Let $R_0 = 5 \cdot 10^{-10}$ and $R_1 = 7.5 \cdot 10^{-10}$, and use the same affine map and center as (3.2). The image of a square in $z$ is a parallelogram in $(k, d)$, not an independently varying rectangle.

For a variable radius $r_* > 0$, retain the exact origin allowance

$$\begin{aligned} e_0(r_*) = {} & 5 \cdot 10^{-21} + E_{60} + 2[(1 - .11r_*)^{-2} - 1 - 2(.11r_*)] \\ & + .12E_{60}r_* + 10^{-20}r_* + 3/2^{256}. \end{aligned} \tag{5.1}$$

Write $b_{\text{boot}}$ for a bootstrap multiplier, distinct from the ODE coefficient $B = 751/4$. Set

$$\begin{aligned} E(r_*, b_{\text{boot}}) &= 8501e_0(r_*) + 580000\,10^{-19} + 4.3(250000)1.001(b_{\text{boot}}r_*)^2, \\ I(r_*, b_{\text{boot}}) &= 4.3(2.70002r_* + E(r_*, b_{\text{boot}})) + \frac{4.3(1.001)(b_{\text{boot}}r_*)^2}{.23^3}. \end{aligned} \tag{5.2}$$

The exact linear bound $2.70002 = 2.7 + .001(.02)$ is retained instead of rounding to 2.72. All central trajectory, denominator and transfer data are independent of the chosen shooting-square radius.

**Proposition 5.1** (Uniform finite-only extension)**.** *Conditional on the source's reference certificate and its analytic enclosures, every parameter in $\|z\|_\infty \le R_0$ has a positive shot on $[0, 56]$ satisfying*

$$|q| < .202R_0, \qquad \|\xi^{\mathbb{R}} - z\|_\infty < .069R_0.$$

*Every parameter in $\|z\|_\infty \le R_1$ has a positive shot on $[0, 56]$ satisfying*

$$|q| < .25R_1, \qquad \|\xi^{\mathbb{R}} - z\|_\infty < .13R_1.$$

*These are two-real-coordinate maximum-norm bounds, not complex-modulus bounds for $\xi - z$.*

*Proof.* For $(r_*, b_{\mathrm{boot}}) = (R_0, 12)$ or $(R_1, 12.5)$, exact rational checks give $b_{\mathrm{boot}} r_* < .23$ and $.23/(.23 - b_{\mathrm{boot}} r_*) < 1.001$. The same nonlinear identity (3.12) yields the endpoint allowance $E$ and partial-block allowance $I$ in (5.2). The fourth component is zero. The initial allowance $.02r_* + e_0(r_*) < b_{\mathrm{boot}} r_*$ and the strict inequalities in the following table close the value bootstrap:

| Radius | $b_{\mathrm{boot}}$ | Upper bound for $E/r_*$ | Upper bound for $I/r_*$ |
|---|---|---|---|
| $R_0$ | 12 | .077593794 | $11.94376479 < 12$ |
| $R_1$ | 12.5 | .126180392 | $12.15270315 < 12.5$ |

The displayed decimals are outward summaries of exact rational values. Positivity and continuation follow by the same denominator and derivative Gronwall argument used in Section 3.4. The origin holomorphic polydisc remains unchanged; its real positivity proof still applies because $d > .4$ and the nonconstant sum before $s = 1$ is at most $.8/3$.

For $R_1$, the exact $d$ interval is

$$[.40068272184853141,\ .40068272186474659] \subset [.40068272184,\ .40068272187].$$

The larger enclosing interval is necessary: the old lower endpoint .40068272185 is insufficient. Exact interval substitution into (4.1) and its $d$ derivatives preserves both coefficient majorant rows in Section 4. For each derivative order $a = 0, 1, 2$, the resulting endpoint-jet bound is

$$\sum_{j=1}^{4} D_j \frac{(3j)(3j+1)\cdots(3j+a-1)}{56^{3j+a}} < .01,$$

where $D = (304, 100000, 65000000, 100000000000)$ and an empty product is one. This justifies the full-segment $P_a(d)$ variation for both radii.

The fixed central residual is still bounded by $.00612R_0$, not $.00612r_*$. Therefore the source projection calculation gives

$$\begin{aligned} |q| &\le .00612R_0 + .1241r_* + .92(E + .00013r_*), \\ \|\xi^{\mathbb{R}} - z\| &\le .00612R_0 + .0091r_* + .68(E + .00013r_*). \end{aligned} \tag{5.3}$$

The .00013 is the sum of transfer error .001(.02) and polynomial variation .01(.011). Exact checks yield respectively upper normalized bounds $(.201725891, .068072180)$ at $R_0$, and $(.244385561, .099071067)$ at $R_1$, proving the claims. □

At $R_1$, the endpoint error exceeds $.11R_1$, so the old $.11R$ bound cannot be used. Similarly the original tail lemma only admits $|q| \le .26R_0$: finite shots in the larger square are not automatically admissible for that lemma. We do not enlarge the tail-matching or global parabolic quantifiers here. The next section returns explicitly to the original $R = R_0$ and the original tail source ball.

# 6 New result: contractive finite-to-tail matching

## 6.1 Precisely delimited input

The next theorem assumes the finite inequalities (3.10) and (4.5), the stated local rounding bounds, and their analytic interpretation from the disk and origin arguments. The already proved source root/tail bounds are also inputs. It does not assume a parameter-dependent interval Hessian calculation, any spectral stability of the shrinker, or a compact PDE solution. The extension is therefore reusable once those same finite inputs have been certified. Its additional arithmetic is small and rational.

**Lemma 6.1** (Origin derivative remainder)**.** *Let $J(z)$ be the exact four-component initial state at $s = 1$ and let $Z$ be the source rounded variation matrix. On $\mathcal{Z}$,*

$$\|DJ(z) - Z\| < 8 \cdot 10^{-11}.$$

*The fourth row of this difference is identically zero.*

*Proof.* For each of the top three entries, the joint holomorphic bound 2 on the polydisc of radius .1 bounds each bivariate Taylor coefficient of total degree $j$ by $2\,10^j$. There are $j+1$ monomials. Each affine parameter displacement and its derivative in a direction $\|v\|_\infty = 1$ are bounded by $.011R$ and .011 respectively. Differentiating the terms of degree at least two therefore costs at most

$$2(.11)\sum_{j\geq 2} j(j+1)(.11R)^{j-1} = .44[(1-.11R)^{-3} - 1]. \tag{6.1}$$

The variation midpoint errors have combined row bound $10^{-20} + 2Q^{-1}$, and their missing analytic tail at most $.12E_{60}$. Adding these terms to (6.1) gives less than $8 \cdot 10^{-11}$. Constant midpoint jumps have zero parameter derivative. The $d$ coordinate is exactly affine. □

**Lemma 6.2** (Differentiated shooting control)**.** *Under the stated source inputs, throughout $\mathcal{Z}$,*

$$\mathrm{Lip}(q_{\mathrm{shot}}) < .31, \qquad \mathrm{Lip}(\xi^{\mathbb{R}}_{\mathrm{shot}} - \mathrm{id}) < .146. \tag{6.2}$$

*Lipschitz constants use the maximum norm on the square and the real two-coordinate output.*

*Proof.* Source continuation first gives $|u| \leq 14R$, independently of any derivative bound. For $a \geq m_j$,

$$N'_a(u) = \frac{u(2a+u)}{a^2(a+u)^2}, \qquad |N'_a(u)| \leq \frac{2.002(14R)}{m_j^3}.$$

The last inequality follows since $14R/.23 < 1$ and $(2 + 14R/.23)/(1 - 14R/.23)^2 < 2.002$. The reference $a$ and its rounded starting data are independent of the shooting parameter.

Differentiate the exact solution in any fixed direction $v$ of norm one. Ordinary smooth dependence holds on a neighborhood of the compact square because the exact shots stay away from $-1$. Let $E_i$ denote the derivative of the endpoint remainder relative to $T_{i,0}Zz$. Temporarily bound the value derivative by 13 inside reached blocks. The derivative of every reference jump is zero. Lemma 6.1, the exact weighted transfer sum, and variation of constants give

$$\begin{aligned}\|E_i\| &\leq 8501(8 \cdot 10^{-11}) + 4.3(250000)2.002(14R)(13) \\ &= .19584633008 < .2.\end{aligned} \tag{6.3}$$

Inside the current block the first-row/third-column bounds yield

$$|\partial_v p| \leq 4.3(2.72 + .2) + \frac{4.3\,2.002(14R)(13)}{.23^3} < 12.556065 < 13. \tag{6.4}$$

The bootstrap starts from $\|Z\| + 8 \cdot 10^{-11} < .021$ and continues through every block. Bounds at square boundary parameters follow by continuity from interior parameters. This proves a uniform derivative bound without differentiating an interval enclosure as if it were the exact solution.

At the endpoint, replacement of $T_{N,0}$ by $\widetilde{T}_{N,0}$ costs .001(.02) in derivative norm, and the subtracted polynomial $P_a(d(z))$ costs at most .01(.011). The constant residual $A_*$ contributes no derivative. Thus

$$\begin{aligned} \|Dq_{\text{shot}}\| &< .116 + .003(2.7) + .92(.2 + .001(.02) + .01(.011)) \\ &= .3082196 < .31, \end{aligned} \tag{6.5}$$

$$\begin{aligned} \|D\xi^{\mathbb{R}}_{\text{shot}} - I\| &< .001 + .003(2.7) + .68(.2 + .001(.02) + .01(.011)) \\ &= .1451884 < .146. \end{aligned} \tag{6.6}$$

Integrate the derivative along segments in the convex square. □

**Lemma 6.3** (Tail dependence on both parameters)**.** *For the selected tails $|q| \le .26R$, $\|j\| \le .3R$, and the source interval of $d$, one has*

$$|\xi(q,d) - \xi(\widetilde{q},\widetilde{d})| \le \frac{.82}{.43}(.36|q-\widetilde{q}| + .0041|d-\widetilde{d}|). \tag{6.7}$$

*Proof.* We must bound $d$ dependence of the residual, not just $P_a(56)$. Let $D_j$ be the four coefficient-derivative bounds in Section 4. Their positive majorants decrease with $s$, so $|\partial_d p_a(s)| < .01$ uniformly on the whole half-line. Directly differentiating the exact residual gives on $s \ge 56$

$$|\partial_d \mathcal{E}_d(p_a)| \le \frac{|C|}{56^3} + \sum_{j=1}^{4} \frac{|L(-3j)|D_j}{56^{3j+3}} + (1-.001)^{-2}\sum_{j=1}^{4}\frac{D_j}{56^{3j}} < .004. \tag{6.8}$$

This deliberately ignores favorable cancellations and is still sufficient. At fixed $q, j$, the Green solution $V$ does not depend on $d$, since $D_0$ is fixed. Also $N''(p) = 2/(1+p)^3 < 2.01$ on $|p| < .001$. Therefore the $d$ derivative of the nonlinear difference in (4.4) is bounded by $2.01(.848R)(.01)$. Its sum with .004 is below .0041.

At fixed $d, j$, a change in $q$ changes the homogeneous Green term; the source bound .36 bounds its effect in (4.4). At fixed $q, d$, changing $j$ costs at most .57. For the two fixed points, subtraction and absorption consequently give

$$\|j - \widetilde{j}\|_\infty \le \frac{.36|q-\widetilde{q}| + .0041|d-\widetilde{d}|}{1-.57}.$$

The nonreal coordinate in (4.3) depends only on the backward convolution of $j$ and has response norm below .82. This proves the result. The mean-value argument stays inside the convex parameter interval and the common source ball. □

**Theorem 6.4** (Contractive matching, conditional on the finite ledger)**.** *Under the source finite inputs specified above, the map* (4.7) *is a contraction on $\mathcal{Z}$, with Lipschitz constant strictly below* .36*. It has exactly one fixed point in that square, and hence exactly one matched shot there with the tail selected from the source ball $\|j\| \le .3R$.*

*Proof.* The source value estimates give $\mathcal{B}(\mathcal{Z}) \subset .38\mathcal{Z}$. By (3.2), $\mathrm{Lip}(d) < .011$. Lemmas 6.2 and 6.3 give

$$\mathrm{Lip}(\mathcal{B}) < .146 + \frac{.82}{.43}(.36(.31) + .0041(.011)) < .358905 < .36. \tag{6.9}$$

The closed square is complete. Banach's theorem gives existence and uniqueness there. The real coordinate and $d$ already agree by the definition of the tail, so a fixed point matches all three jet entries. The affine map in (3.2) has nonzero determinant, so uniqueness in $z$ is uniqueness of $(k, d)$ within its image. No conclusion is made about tails outside the selected small ball or shots outside this square. □

**Corollary 6.5** (Residual, iteration, and perturbation bounds)**.** *Let $z_*$ be this fixed point. For every $z \in \mathcal{Z}$,*

$$\|z - z_*\| \le \frac{25}{16}\|z - \mathcal{B}(z)\|. \tag{6.10}$$

*Exact Picard iterates satisfy $\|z_j - z_*\| \le .36^j\|z_0 - z_*\|$ and $\|z_j - z_*\| \le .36^j\|z_1 - z_0\|/.64$. If a perturbed map has a fixed point $\widehat{z} \in \mathcal{Z}$ and $\sup_{\mathcal{Z}}\|\widehat{\mathcal{B}} - \mathcal{B}\| \le \varepsilon$, then $\|\widehat{z} - z_*\| \le 25\varepsilon/16$.*

*Proof.* Add and subtract $\mathcal{B}(z)$ and use the contraction. The iteration estimate follows by induction, and the a posteriori successive-difference bound follows by summing a geometric series. For the perturbed fixed point apply the same subtraction at $\widehat{z}$. Existence of that fixed point is an assumption, not a consequence of the error estimate alone. □

For a continuous perturbation with $\varepsilon \le .62R$, the source $.38R$ image margin also supplies a Brouwer self-map and hence existence. It need not give uniqueness unless an independent contraction bound is retained. Provided the current iterate $z_j$ remains in $\mathcal{Z}$, any computed map evaluation error $\varepsilon_j$ enters $\|z_{j+1} - z_*\| \le .36\|z_j - z_*\| + \varepsilon_j$. Repeated use requires each current iterate to remain in this square; the inexact iteration is not asserted to preserve it automatically. These are accuracy statements, not a measured speedup: validated evaluation of a tail on a half-line still has a cost.

The physical parameter and curvature-coefficient consequences are particularly explicit:

$$\begin{aligned} |\Delta k| &\le .00088921\,\|\Delta z\|, & |\Delta d| &\le .01081012\,\|\Delta z\|, \\ |\Delta c| &\le \frac{.01081012}{(1+d_-)^2}\|\Delta z\|, & |\Delta a_{\text{curvature}}| &\le \frac{110\sqrt{27}}{4}(.00088921)\|\Delta z\|, \end{aligned} \tag{6.11}$$

where $d_- = d_0 - 5.40506 \cdot 10^{-12} > 0$. These concern the matched profile parameters. They do not bound the lifespan or distance between the compact metrics produced by the separate gluing construction.

# 7 Ancient, parabolic, and compact dependencies

## 7.1 Weights and what a finite inverse does not supply

For $r = \sqrt{x}$, $\sigma = (r^2 + \rho)^{1/2}$, and $p_0 = -2 + 2\delta$, the source spaces use local units $(\sigma, \sigma^4)$. $X$ controls four spatial derivatives and one time derivative, all with parabolic $C^\alpha$ seminorms, weight $p_0$. $X'$ omits only the time derivative, not temporal seminorms of spatial derivatives. $Y$ has weight $p_0 - 4$. At the center, $|V(t,0)| \lesssim \|V\|_{X'}\rho^{-1+\delta}$.

Uniform local Schauder estimates have to remain valid down to a zero-data initial face as the time length shrinks. The frozen fourth-order kernel has $|D^j k_b(z)| \lesssim b^{-(m+j)/4}(1 + |z|b^{-1/4})^{-N}$; spatial cancellation gives $\|D^4 k_b * f\| \lesssim [f]_\alpha b^{-1+\alpha/4}$. Splitting time lags at the spatial increment to the fourth power, or at the temporal increment, proves the parabolic estimate. Localization retains coefficient derivatives, and spatial interpolation absorbs the lower-order terms without an inverse power of the elapsed time.

The source zero-data outer estimate is

$$\|u\|_X \le C\|f\|_Y, \quad \sup s^{-p_0}|u| \le C\eta\|f\|_Y, \quad \eta = T/(\inf s)^4.$$

Lower spatial derivatives gain $\eta^{(4-j-\alpha)/4}$ for $j \le 3$; the full $X'$ norm gains $\eta^{(\alpha'-\alpha)/4}$ only with a source exponent $\alpha' > \alpha$. One cannot make $u_t$ small because $u_t(0) = f(0)$, nor drop the exponent gap: the circle example $u_N = N^{-4-\alpha}(1 - e^{-N^4 t})\cos Nz$ on $[0, N^{-4}]$ has a fourth-derivative $C^\alpha$ seminorm of order one.

### 7.2 The ancient right inverse is a spectral construction

The flat linearization is

$$\widehat{\mathcal{P}}V = V_t + \frac{G(|z|^2/\rho)^2}{16}\Delta_z^2 V + 2|z|^2 G(|z|^2/\rho)\widehat{\mathcal{A}}F\,V.$$

In similarity variables $\tau = -\log(-t)$, $\zeta = z/\sqrt{\rho}$, $V = \rho^{-1}v$, the equation is

$$v_\tau + \frac{A_f}{16}\Delta^2 v + \tfrac{1}{4}\zeta\cdot\nabla v + (\tfrac{1}{2} + B_f)v = f^{\#}, \quad A_f = G(|\zeta|^2)^2, \quad B_f = -yG'(y)/G(y).$$

The all-order tail expansion gives derivatives of $A_f - c^2$ and $B_f$ decaying like $(1+|\zeta|)^{-4-|\nu|}$. This is where a merely undifferentiated finite tail estimate would fail to suffice.

The source constructs the evolution on radial $C_0(\mathbb{R}^{24})$ via moving coordinates, a frozen-coefficient parametrix, and a Volterra residual of order $(s-r)^{-3/4}$. Iterated kernels have beta-integral coefficients $C^k\Gamma(1/4)^k s^{k/4-1}/\Gamma(k/4)$, giving a convergent series. Weighted $L^2$ energy gives uniqueness. To prove compactness of the time-one difference $P - P_c$, spatially translated countersequences converge to the same bounded constant-coefficient evolution with the same distributional initial trace. Uniqueness of that limit supplies uniform small spatial tails; local equicontinuity then gives compactness. Compactness is not obtained by simply calling a principal-coefficient perturbation lower order.

The constant evolution has $\|P_c^j\| \le Ce^{-j/2}$. Analytic Fredholm theory isolates finite-multiplicity spectrum outside $e^{-1/2}$. Choose a circle $q \in (e^{-1/2}, 1)$ avoiding the spectrum, and put $\delta = -2\log q \in (0,1)$. Riesz projections give $E_s \oplus E_u$ with powers bounded by $Cr_s^j$ and inverse powers by $Cr_u^{-j}$, where $r_s < q < r_u$ and $E_u$ is finite-dimensional. For source increments $l_j$ the two-sided recursion is

$$v_j = \sum_{k\ge 0} P_s^k \Pi_s l_{j-1-k} - \sum_{k\ge 0} P_u^{-k-1}\Pi_u l_{j+k}.$$

The future-forcing sum selects initial data; it is not a forward stability theorem for arbitrary prescribed metrics. The preliminary bound $|V| \lesssim \rho^{-1+\delta}$ is upgraded on outer rescaled annuli by $\int_0^4 q^{-1+\delta}\,\mathrm{d}q = 4^\delta/\delta < \infty$ and an interior $L^2$ Schauder estimate. The strict lower bound $\delta > 0$ matters. No numerical value of $\delta$, inverse norm, or unstable dimension is extracted here from the finite shooting map.

### 7.3 Compact gluing retains exact commutator cancellation

Write $g_c = 1 + v_* F$, $g_f = 1 + xF$, and $\mathcal{P} = \partial_t + g_c^2\mathcal{A} + 2v_* g_c \mathcal{A}F$. In core coordinates

$$\Delta_\Sigma = \Delta_z - z_i z_j \partial_{ij} - 29 z_i \partial_i, \qquad \mathcal{A} = \Delta_\Sigma^2/16 - (37/4)\Delta_\Sigma + 312.$$

At scale $b = \sigma$, the flat/compact coefficient mismatch has scaled size $O(b^2)$ and the residual $E_c = F_t + g_c^2\mathcal{A}F$ has size $O(b^{-4})$, hence $Y$ size $O(b^{2-2\delta})$. The outer expansion $F = (c-1)r^{-2} + \sum a_j(-t)^j r^{-4j-2}$ with differentiated remainders supplies smoothness to $t = 0$ away from the core.

For $\chi = 1$ on $[0,1]$, $\chi = 0$ on $[2,\infty)$, the four cutoffs are $\chi_i = \chi(r/\beta)$, $\chi_b = \chi(r/(2\beta))$, $\chi_o = 1 - \chi(2r/\beta)$, $\kappa = 1 - \chi(4r/\beta)$. The inner source is extended below $-T$ at local time scale $(|z|^2 + \sqrt{T})^2$, not by an uncontrolled discontinuous extension. The ancient inverse gives $V_i = \chi_b \widehat{D}\mathcal{E}_i f$ and

$$\mathcal{P}V_i = \chi_i f + R_i f + J_i f, \quad \|R_i\| \lesssim \beta^2, \quad \|J_i\| \lesssim 1.$$

The non-small $J_i$ is canceled exactly by the outer forcing $f_o = f - \chi_i f - J_i f$. The outer coefficient modification equals the compact operator wherever $\chi_o$ is nonzero. Its zero-data solution $Z_o$ gives $V_o = \chi_o Z_o$ and

$$\mathcal{P}(V_i + V_o) = f + (R_i + J_o)f, \quad \|J_o\| \lesssim \eta^{(1-\alpha)/4}, \quad \eta = T/\beta^4.$$

Consequently $D = D_{\mathrm{app}}(I + \mathcal{R})^{-1}$ is a right inverse when $\|\mathcal{R}\| < 1$, not an assertion of injectivity.

For the particular residual, split with the *output* cutoff $E_c = \chi_b E_c + (1 - \chi_b)E_c$. The second term has zero inner input exactly. The identity $Df_e = D_{\mathrm{app}} f_e - D\mathcal{R} f_e$ yields

$$\|DE_c\|_{X'} \lesssim \beta^{2-2\delta} + \eta^{(\alpha'-\alpha)/4} + \eta^{(1-\alpha)/4}. \tag{7.1}$$

This is stronger than boundedness of $D$ on a bounded residual. The nonlinearity is

$$N(V) = v_*^2 V^2 \mathcal{A}F + (2v_* g_c V + v_*^2 V^2)\mathcal{A}V.$$

It maps an $X'$ ball of radius $\varepsilon$ to $Y$ with size $C(\varepsilon^2 + \varepsilon^3)$ and Lipschitz constant $C(\varepsilon + \varepsilon^2)$. Fix the uniform inverse and nonlinear constants first, then the radius, then $\beta$, then $\eta$. The contraction $V = -DE_c - DN(V)$ is performed in $X'$; its fixed-point identity subsequently gives $V \in X$ and the missing time derivative. Positivity follows from (2.7) and $|v_* V| \lesssim \varepsilon\sigma^{2\delta}$, and parabolic bootstrapping gives smoothness at interior negative times.

These are the source's analytic dependencies. The new shooting contraction does not replace any of them. Products with a stationary cscK factor inherit the blowup because scalar curvatures and averages add; for $\mathbb{CP}^{10} \times \mathbb{CP}^m$ the factor $(m+1)\omega_{\mathrm{FS,m}}/11$ makes the reference product Einstein. This does not construct a metric on $\mathbb{CP}^{10+m}$.

# 8 Tests, literature, limitations, and next steps

## 8.1 Bounded execution contract

The author check program passed 70 exact rational inequality and coefficient-identity checks. Descriptive decimal outputs are not interval evidence. It checks the new derivative remainder, nonlinear derivative factor, endpoint and interior derivative bootstrap, residual $d$ derivative, and contraction constant. Independent coefficient tests check the reciprocal/tangent identities and the scaled ODE coefficient update. These tests do not independently construct all 288 block transfers. The inspected pinned upstream scripts `certificate.py`, `arithmetic_allowances.py`, and `tail-check.py` all exited with code zero and assertions enabled. The main certificate runtime was 121.609 seconds. A separate finite-refinement script passed 26 checks; an independent matching-derivative reconstruction passed 14 checks. A further independent matching review passed 22 rational checks and found no unresolved high- or medium-priority defect in the derivative, norm-conversion, continuation, or selected-tail-branch arguments. These records distinguish upstream replay, new scalar/interval budget checks, and mathematical review of their analytic interpretation. No computation of the ancient spectrum, compact inverse norm, gluing time, or a particular initial metric is claimed.

**Independent global review.** A separate bounded audit read all source sections and found no unresolved high- or medium-priority mathematical defect in the reviewed global chain. It checked the future-forcing choice, annular $L^2$ upgrade, exponent gap, commutator cancellation, dynamic affine normalization, and fixed-point curvature obstruction. This is not formal verification or an exhaustive independent proof. The cited Kato chapter text was not directly inspected by that auditor; standard spectral facts remain imports. Public import checks and their depths are recorded separately from the manuscript and certificate byte comparisons.

### 8.2 Public context checked without private queries

Public queries used the problem title and public paper URLs only. Current arXiv metadata checked on October 7–8, 2026 identifies *On the Calabi flow* as version 3 (October 2007), which describes short-time theory, stability near cscK metrics, and extension under Ricci bounds [2]. These restricted statements are not arbitrary-data smooth global existence. The checked $L^p$ paper is version 2 (September 2024), stating extension for uniformly bounded spatial scalar-curvature $L^p$ norm with $p > n$ [3]; its sequel is version 2 (November 2025), with space-time integral criteria [4]. The publisher text of Berman–Darvas–Lu, Theorem 1.11, gives weak-flow $d_1$ convergence to a smooth cscK potential when one exists [5]. That asymptotic weak topology does not imply smooth continuation through every finite time. No absence of later work or historical priority is inferred from a title search. The source remains a pinned preprint claim, not external peer-review certification.

Li–Wang–Zheng's October 4, 2026 preprint, Theorem 1.2, states finite-endpoint continuation for compact Kähler surfaces (complex dimension two) under uniformly bounded Sobolev constants [6]. The bounded check covered its introduction, theorem, Corollary 1.3, and convention (2.3), not its full proof. Neither the dimension nor the additional hypothesis supplies a contradiction to the claimed $\mathbb{CP}^{10}$ example or a higher-dimensional extension criterion.

### 8.3 Benefits, costs, and research boundaries

The realized benefit is qualitative as well as quantitative: the same finite source ledger supports a unique matched parameter in a stated branch instead of mere Brouwer existence, with a computable residual amplification constant. The private residual-ball and perturbation principles [8] guide the proof, but the numerical derivative estimates and the $d$-dependent tail residual are specific to this source. The exact reciprocal and tangent recurrences are inherited, not rebranded as new.

The main cost is conservative norm propagation. The constant .36 is not optimized; the estimates retain 8501, 250000, and $14R$ instead of recomputing tighter transfer information. This avoids extra large computation but does not establish faster verification. Uniform control on a square of radius $5 \cdot 10^{-10}$ is a narrow local statement. Tail uniqueness holds in the selected source ball, and parameter uniqueness does not imply dynamical stability or uniqueness of the compact initial metric.

Concrete next steps are: certify second parameter variations if a larger shooting neighborhood is desired; use row-specific transfer actions to tighten the matching contraction without discarding rounding terms; give a validated finite representation of the tail Green integrals for actual Picard evaluation; and separately quantify the ancient spectral circle and inverse constants before claiming a constructive gluing time. A lower-dimensional profile would require a new origin/tail matching analysis and finite ledger, not substitution of a different $n$ into the existing numbers. Stability under arbitrary metric perturbations remains a research question, not a consequence of this finite certificate.

## A Dependency and provenance record

| Claim | Required input and status |
|---|---|
| Fixed-class reduction | Smooth positive $H$, endpoint reconstruction, affine Legendre prescription; source derivation reproduced in Section 2. |
| Positive flat profile | Origin holomorphy, finite ledger, root bounds, tail contraction, matching, differentiated asymptotics; imported source construction, not implied by recurrence identities alone. |

| Claim | Required input and status |
|---|---|
| Finite-box extension | Recomputed radius-dependent origin/nonlinear budgets, fixed $R_0$ central projection and widened $d$ interval; Proposition 5.1, exact checks and independent review. |
| Contractive matching | Same source ledger plus Lemmas 6.1–6.3; conditional extension proved here with bounded rational checks. |
| Ancient right inverse | Whole-space evolution, compact time-one perturbation, spectral circle, two-sided recursion, annular $L^2$ upgrade; independent infinite-dimensional source argument. |
| Compact realization | Uniform local estimates, exact commutator cancellation, right-inverse Neumann series, special residual smallness, nonlinear $X'$ contraction; separate source argument. |
| Finite-time singularity | Compact realization, $G'(0) < 0$, fixed-point curvature identity, smooth uniqueness; imported headline claim. |
| Executed evidence | Author rational/identity checks and parent results as recorded, with locators and hashes in the final manifest. No unrecorded full verifier or geometric test is implied. |

The final nested path/SHA-256 manifest includes the exact source and private input files, read locators, public-check notes, authored test code and results, parent review evidence, native compiler/open result, MiKTeX export logs and QA images, final TeX and disk PDF, and the contribution record. It is checked by the project's `check_freeze_manifest.py`. The manifest cannot include its own SHA-256 recursively; a separate checksum identifies it. The native editor open request was queued, while native compilation failed with the platform diagnostic "Unable to find standard directories for platform." The disk PDF was exported through installed MiKTeX and visually audited separately; native success is not implied by that export.

## References


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[9] M. Subbarao. *Adaptive Local–Global Graph Operators with Certified Updates and Reconstruction*. Technical Report TR-core-pat-5, October 1, 2026. Confidential local manuscript, Theorems `thm:neumann`, `thm:frame`, `thm:woodbury`, and Proposition `prop:schurresidual`.